
Problem Set 8

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PY 451 Section A1

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1 Commutators and Position Space

1. (a) By the canonical commutation relation, it follows that

$$XP - PX = i\hbar \mathbb{1} \implies PX = XP - i\hbar \mathbb{1}.$$

Since $\mathbb{1} = X^0 P^0$, this suffices as a solution.

(b)

$$\begin{aligned} P^2 X &= P(XP - i\hbar \mathbb{1}) = (PX)P - i\hbar P = (XP - i\hbar \mathbb{1})P - i\hbar P \\ &= XP^2 - 2i\hbar P. \end{aligned}$$

Since $P = X^0 P$, this suffices as a solution.

(c)

$$\begin{aligned} PXPX &= (XP - i\hbar \mathbb{1})PX \\ &= XP^2X - i\hbar PX \\ &= X(XP^2 - 2i\hbar P) - i\hbar(XP - i\hbar \mathbb{1}) \\ &= X^2P^2 - 2i\hbar XP - i\hbar XP - \hbar^2 \mathbb{1} \\ &= X^2P^2 - 3i\hbar XP - \hbar^2 \mathbb{1}. \end{aligned}$$

2.

$$[P, H] = \left[P, \frac{1}{2m} P^2 \right] + [P, V(X)] = [P, V(X)].$$

Since $V(X)$ is a function of X , by identity,

$$[P, V(X)] = -[V(X), P] = -\frac{dV}{dX} [X, P] = -i\hbar \frac{dV}{dX}.$$

In class, we showed that

$$\frac{d}{dt} \langle \mathcal{O} \rangle = \frac{1}{i\hbar} \langle [\mathcal{O}, H] \rangle$$

for $\mathcal{O}$ any time-independent hermitian operator. Plugging in P for $\mathcal{O}$ gives us

$$\frac{d}{dt} \langle P \rangle = \left\langle -\frac{d}{dX} V(X) \right\rangle := \langle F(X) \rangle.$$

3. In finite-dimensional space, a Hermitian operator was an operator that was its own hermitian conjugate. This generalizes to any arbitrary dimensional Hilbert space $\mathcal{H}$ (and further to Banach spaces I believe) by defining a Hermitian operator as an operator that is self-adjoint. More precisely, the Hermitian adjoint $\mathcal{O}^\dagger$ of an operator $\mathcal{O}$ is the unique operator satisfying

$(\langle v|\mathcal{O}^\dagger)|w\rangle = \langle v|(\mathcal{O}|w\rangle)$ for all $|v\rangle, |w\rangle \in \mathcal{H}$. This is one of the few cases where I believe the math notation for inner product is more straightforward: $\langle \mathcal{O}v, w\rangle = \langle v, \mathcal{O}^\dagger w\rangle$. A Hermitian operator is then any operator $\mathcal{O} \in \mathcal{H}^*$ such that $\mathcal{O} = \mathcal{O}^\dagger$.

To show that P is a Hermitian operator, recall that every state $|\psi_1\rangle, |\psi_2\rangle$ can be viewed in position space as a wavefunction $\psi_1(x), \psi_2(x)$. By definition of an adjoint,

$$\begin{aligned}\langle \psi_1|P|\psi_2\rangle &= -i\hbar \int_{-\infty}^{\infty} \psi_1^*(x) \frac{d}{dx} \psi_2(x) dx \\ &= -i\hbar \left([\psi_1^*(x)\psi_2(x)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \left(\frac{d}{dx} \psi_1^*(x) \right) \psi_2(x) dx \right) \\ &= -i\hbar [\psi_1^*(x)\psi_2(x)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} i\hbar \left(\frac{d}{dx} \psi_1^*(x) \right) \psi_2(x) dx \\ &= -i\hbar [\psi_1^*(x)\psi_2(x)]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \left(P^\dagger \psi_1(x) \right)^* \psi_2(x) dx.\end{aligned}$$

Assume the boundary terms vanish at infinity for physically valid wavefunctions. For this equality to hold, we must have that $P^\dagger = -i\hbar \frac{d}{dx} = P$.

4. Since the adjoint distributes linearly,

$$A^\dagger = P^\dagger + c^*W(X)^\dagger = P + c^*W(X).$$

We thus have

$$\begin{aligned}[A^\dagger, A] &= [P + c^*W(X), A] \\ &= [P, A] + [c^*W(X), A] \\ &= [P, P + cW(X)] + [c^*W(X), P + cW(X)] \\ &= [P, P] + [P, cW(X)] + [c^*W(X), P] + [c^*W(X), cW(X)]\end{aligned}$$

I will compute each commutator individually.

$$[c^*W(X), cW(X)] = c^*cW(X)^2 - cc^*W(X)^2 = W(X)^2 (|c|^2 - |c|^2) = 0.$$

$$\begin{aligned}[P, cW(X)] + [c^*W(X), P] &= cPW(X) - cW(X)P + c^*PW(X) - c^*W(X)P \\ &= PW(X)(c + c^*) - W(X)P(c + c^*) \\ &= (\text{Re } c)(PW(X) - W(X)P) \\ &= (\text{Re } c)[P, W(X)].\end{aligned}$$

Therefore,

$$[A^\dagger, A] = (\text{Re } c)[P, W(X)].$$

This can be computed further by recalling the identity $[P, W(X)] = -\frac{dW}{dX}[X, P] = -i\hbar \frac{dW}{dX}$. Therefore, we also have

$$[A^\dagger, A] = -i\hbar(\text{Re } c) \frac{dW}{dX}.$$

2 Infinite Square Well

1. Eigenfunctions of H in the infinite square well take the form

$$\psi_n(x) \sqrt{\frac{2}{L}} \sin(k_n x),$$

where

$$k_n = \frac{n\pi}{L}, \quad E_n = \frac{n^2 \hbar^2 \pi^2}{2mL^2}.$$

Therefore, there exists a collection $\{C_n\}_{n=1}^{\infty} \subseteq \mathbb{R}$ satisfying

$$\psi(x, 0) = \sum_{n=1}^{\infty} C_n \psi_n(x).$$

Although the overall wavefunction $\psi(x, t)$ is not separable, each $\psi_n(x)$ is separable, and thus we can write $\psi_n(x, t) = \phi_n(t) \psi_n(x)$ for some function $\phi_n(t)$. The Schrodinger equation gives

$$\begin{aligned} i\hbar \psi_n(x) \frac{\partial \phi_n}{\partial t} &= \frac{-\hbar^2}{2m} \phi_n(t) \frac{\partial^2 \psi_n}{\partial x^2} \\ \implies i\hbar \frac{1}{\phi_n(t)} \frac{\partial \phi_n}{\partial t} &= -\frac{\hbar^2}{2m} \frac{1}{\psi_n(x)} \frac{\partial^2 \psi_n}{\partial x^2}. \end{aligned}$$

Solving the right side of the equation gives

$$\begin{aligned} -\frac{\hbar^2}{2m} \frac{1}{\psi_n(x)} \frac{\partial^2 \psi_n}{\partial x^2} &= -\frac{\hbar^2}{2m} \frac{1}{\sin(k_n x)} (-k_n^2 \sin(k_n x)) \\ &= \frac{\hbar^2 k_n^2}{2m} = \frac{\hbar^2 n^2 \pi^2}{2mL^2} = E_n. \end{aligned}$$

Therefore,

$$\frac{\partial \phi_n}{\partial t} = -\frac{i}{\hbar} E_n \implies \phi_n(t) = \exp\left(-\frac{iE_n t}{\hbar}\right).$$

We thus have

$$\psi(x, t) = \sum_{n=1}^{\infty} C_n e^{-iE_n t/\hbar} \sin(k_n x).$$

It thus suffices to solve for C_n to complete the problem. It's well known that the ψ_n s are orthogonal, and moreover the time-dependence goes to 1 as t goes to 0, so we obtain

$$\begin{aligned} \int_0^L \psi_m^*(x) \psi(x, 0) dx &= \int_0^L \psi_m^* \sum_{n=1}^{\infty} C_n \psi_n(x) dx \\ &= \sum_{n=1}^{\infty} \int_0^L C_n \psi_m^*(x) \psi_n(x) dx \\ &= \int_0^L C_n |\psi_m(x)|^2 dx \\ &= C_n. \end{aligned}$$

We can explicitly compute each C_n , making heavy use of the fact that $\sin(k_n x) = 0$ and $\psi(x, 0) = 0$ for $x = 0$ or L to make most boundary terms vanish.

$$\begin{aligned}
C_n &= \int_0^L \psi_n^*(x) \psi(x, 0) dx \\
&= \int_0^L \left(\sqrt{\frac{2}{L}} \sin(k_n x) \right) A \left(\left(\frac{x}{L} \right)^3 - \frac{3}{2} \left(\frac{x}{L} \right)^2 + \frac{1}{2} \left(\frac{x}{L} \right) \right) dx \\
&= \int_0^L \sqrt{\frac{2}{L}} \sin(k_n x) A \left(\frac{1}{L^3} x^3 - \frac{3}{2L^2} x^2 + \frac{1}{2L} x \right) dx \\
&= \left[\psi(x, 0) \left(-\frac{1}{k_n} \sqrt{\frac{2}{L}} \cos(k_n x) \right) \right]_0^L + \int_0^L \frac{A}{k_n} \sqrt{\frac{2}{L}} \cos(k_n x) \left(\frac{3}{L^3} x^2 - \frac{6}{2L^2} x + \frac{1}{2L} \right) dx \\
&= \left[\frac{A}{k_n^2} \left(\frac{3}{L^3} x^2 - \frac{6}{2L^2} x + \frac{1}{2L} \right) \psi_n(x) \right]_0^L - \int_0^L \frac{A}{k_n^2} \sqrt{\frac{2}{L}} \sin(k_n x) \left(\frac{6}{L^3} x - \frac{6}{2L^2} \right) dx \\
&= \left[\frac{A}{k_n^3} \sqrt{\frac{2}{L}} \cos(k_n x) \left(\frac{6}{L^3} x - \frac{3}{L^2} \right) \right]_0^L - \int_0^L \frac{A}{k_n^3} \sqrt{\frac{2}{L}} \cos(k_n x) \left(\frac{6}{L^3} \right) dx \\
&= \left[\frac{A}{k_n^3} \sqrt{\frac{2}{L}} \cos(k_n x) \left(\frac{6}{L^3} x - \frac{3}{L^2} \right) \right]_0^L - \frac{6A}{L^3 k_n^4} \sqrt{\frac{2}{L}} [\sin(k_n x)]_0^L \\
&= \left[\frac{A}{k_n^3} \sqrt{\frac{2}{L}} \cos(k_n x) \left(\frac{6}{L^3} x - \frac{3}{L^2} \right) \right]_0^L \\
&= \frac{A}{k_n^3} \sqrt{\frac{2}{L}} \left(\cos(n\pi) \left(\frac{6}{L^2} - \frac{3}{L^2} \right) - \left(-\frac{3}{L^2} \right) \right) \\
&= \frac{A}{k_n^3} \sqrt{\frac{2}{L}} \left(\frac{3(-1)^n}{L^2} + \frac{3}{L^2} \right) \\
&= \frac{6A}{k_n^3 L^2} \sqrt{\frac{2}{L}}, \quad n \text{ even}; \quad 0, \quad n \text{ odd}.
\end{aligned}$$

2. By the previous problem, if n is odd, then the probability is 0. For n even, we compute

$$\begin{aligned}
|\langle E_n | \psi(x, 0) \rangle|^2 &\cong \left| \int_0^L \psi_n^*(x) \psi(x, 0) dx \right|^2 \\
&= C_n^2 \\
&= \frac{72A^2}{k_n^6 L^5} \\
&= \frac{76A^2 L}{n^6 \pi^6}.
\end{aligned}$$

3. The value of the summation came from google.

$$\begin{aligned}
\langle \psi | H | \psi \rangle &= \int_0^L \left(\sum_{n=1}^{\infty} C_n \psi_n(x) \right) \left(\sum_{m=1}^{\infty} C_m H \psi_m(x) \right) dx \\
&= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \int_0^L C_n C_m E_m \psi_n^*(x) \psi_m(x) dx \\
&= \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} C_n C_m E_m \delta_{nm} \\
&= \sum_{n=1}^{\infty} C_n^2 E_n \\
&= \sum_{n=1}^{\infty} \frac{76A^2 L}{(2n)^6 \pi^6} \frac{n^2 \hbar^2 \pi^2}{2mL} \\
&= \sum_{n=1}^{\infty} \frac{76A^2 \hbar^2}{2m(2n)^4 \pi^4} \\
&= \frac{76A^2 \hbar^2}{32m\pi^4} \sum_{n=1}^{\infty} \frac{1}{n^4} \\
&= \frac{2A^2 \hbar^2}{\pi^4 m} \sum_{n=1}^{\infty} \frac{1}{n^4} \\
&= \frac{2A^2 \hbar^2}{\pi^4 m} \frac{\pi^4}{90} \\
&= \frac{A^2 \hbar^2}{45m}.
\end{aligned}$$

4. The ground state of a square well is given by

$$\psi_1(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{\pi x}{L}\right).$$

The ground state of the new square well is given by

$$\varphi_1(x) = \sqrt{\frac{2}{3L}} \sin\left(\frac{\pi x}{3L}\right).$$

The first excited state of the new well is given by

$$\varphi_2(x) = \sqrt{\frac{2}{3L}} \sin\left(\frac{2\pi x}{3L}\right).$$

The probability $\mathcal{P}_1$ of finding the ground state is computed as follows. Since $\psi_1(x)$ is 0 outside

of $(0, L)$, we only compute the integral over that interval.

$$\begin{aligned}
\mathcal{P}_1 &= \left| \int_0^L \varphi_1^*(x) \psi_1(x) dx \right|^2 \\
&= \left| \int_0^L \frac{2}{L\sqrt{3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right|^2 \\
&= \frac{4}{3L^2} \left| -\left[\frac{3L}{\pi} \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{3L}\right) \right]_0^L + \int_0^L 3 \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{\pi x}{3L}\right) dx \right|^2 \\
&= \frac{4}{3L^2} \left| \left[\frac{9L}{\pi} \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) \right]_0^L + \int_0^L 9 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right|^2 \\
&= \frac{4}{3L^2} \left| -\frac{9L\sqrt{3}}{2} + \int_0^L 9 \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right|^2 \\
&= \frac{4}{3L^2} \left(\left(-\frac{9L\sqrt{3}}{2\pi} \right)^2 + 81 \left| \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right|^2 - \frac{81L\sqrt{3}}{\pi} \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right) \\
&= \frac{4}{3L^2} \left(\frac{243L^2}{4\pi^2} - \frac{81L\sqrt{3}}{\pi} \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx \right) + 81\mathcal{P}_1.
\end{aligned}$$

It's easy to see from the work here that

$$9 \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx = \frac{9L\sqrt{3}}{2\pi} \implies \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{\pi x}{3L}\right) dx = \frac{L\sqrt{3}}{2\pi}.$$

We thus have

$$\mathcal{P}_1 = \frac{4}{240L^2} \left(\frac{243L^2}{2\pi^2} - \frac{243L^2}{4\pi^2} \right) = \frac{243}{240\pi^2}.$$

I refuse to simplify these numbers.

Calculating $\mathcal{P}_2$ is similar. I will do this integral in a smarter way that will probably prevent algebra errors. First, we compute the integral

$$\begin{aligned}
\int_0^L \varphi_2^*(x) \psi_1(x) dx &= \int_0^L \frac{2}{L\sqrt{3}} \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{3L}\right) dx \\
&= \frac{2}{L\sqrt{3}} \left(-\left[\frac{3L}{2\pi} \sin\left(\frac{\pi x}{L}\right) \cos\left(\frac{2\pi x}{3L}\right) \right]_0^L + \int_0^L \frac{3}{2} \cos\left(\frac{\pi x}{L}\right) \cos\left(\frac{2\pi x}{3L}\right) dx \right) \\
&= \frac{2}{L\sqrt{3}} \left(\left[\frac{9L}{4\pi} \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{3L}\right) \right]_0^L + \frac{9}{4} \int_0^L \sin\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{3L}\right) dx \right).
\end{aligned}$$

It follows that

$$-\frac{5}{4} \int_0^L \varphi_2^*(x) \psi_1(x) dx = \frac{2}{L\sqrt{3}} \left[\frac{9L}{4\pi} \cos\left(\frac{\pi x}{L}\right) \sin\left(\frac{2\pi x}{3L}\right) \right]_0^L = -\frac{2}{L\sqrt{3}} \frac{9L}{4\pi} \left(\frac{\sqrt{3}}{2} \right).$$

Therefore,

$$\int_0^L \varphi_2^*(x) \psi_1(x) dx = \frac{9\sqrt{3}}{5\pi}.$$

The probability is the absolute value squared of this, so

$$\mathcal{P}_2 = \frac{243}{25\pi^2}.$$